Question #1 of 120

Question ID: 1214666

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Which of the following statements about ethical behavior is *most accurate*?

A) Legal standards typically set a higher standard of behavior than ethical standards.

B) Personal traits are the most important determinant of the ethical quality of one's behavior.

C) **Unethical behavior in the investment industry increases the cost of capital for businesses.**

Explanation

Unethical behavior in the investment industry increases the cost of capital for businesses by increasing savers' perceived risk of providing funds, causing them to demand a higher rate of return on the capital they provide. Situational influences external to an individual are claimed to be a more important determinant of ethical behavior than personal traits. Ethical standards often set a higher standard of behavior than legal standards.

For Further Reference:

(Study Session 1, Module 1.1, LOS 1.d, 1.f, 1.g)

CFA® Program Curriculum, Volume 1, page 15

CFA® Program Curriculum, Volume 1, page 12

CFA® Program Curriculum, Volume 1, page 17

Question #2 of 120

Question ID: 1214672

While at dinner a member overhears a well-known and influential analyst from a different firm tell his companion that he will be raising his recommendation on Tree Products from hold to buy when he appears on a morning talk show the next day. Before the broadcast, the member buys 1,000 shares of Tree Products for her own account. The member has violated the Standard relating to:

A) fair dealing.

B) diligence and reasonable basis.

C) **material nonpublic information.**

Explanation

A change in investment recommendation by a well-known and influential analyst is likely to affect the market price and is considered material information, so acting on this information before it has been released is prohibited by Standard II(A) Material Nonpublic

Information. Because the member does not work for the analyst's firm, this is not addressed by Standard III(B) Fair Dealing. Having a reasonable basis does not apply to a member's own personal trades.

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #3 of 120

Question ID: 1214668

Allen Winkler, CFA, recently had lunch with Kim Thompson, a former professor of his, who told him of a new valuation model she had developed. Winkler recreated Thompson's model with some revisions and back-tested it using data provided by Standard & Poor's (S&P) with impressive results. Winkler's firm launches a mutual fund based on the revised model, and Winkler provides a discussion of the principles underlying the model and the test results. Is Winkler required to credit Thompson for having developed the model and S&P as the source of the data?

A) Both of these sources must be cited.
B) Neither of these sources must be cited.
C) **Only one of these sources must be cited.**

Explanation

Under Standard I(C) Misrepresentation, Winkler must identify Thompson as having developed the original model to avoid the prohibition against plagiarism. The only permitted exception is using factual information published by recognized financial and statistical reporting services such as S&P.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #4 of 120

Question ID: 1214679

Donald Smith, CFA, has been assigned by his employer to write a report for clients on Braden Corporation. Smith has 1,000 shares of Braden that he bought three years ago and has been discussing a consulting contract with Braden to write guidelines for their investor relations department. If Smith writes the report on Braden Corporation, he must disclose within the report:

A) **both his ownership of Braden shares and his prospective consulting work.**
B) and to his employer his prospective consulting work but not his ownership of Braden shares.
C) his ownership of Braden shares but need only disclose his prospective consulting work to his employer.

Explanation

Both ownership of Braden stock and the possible consulting work present potential conflicts of interest for Smith and must be disclosed within the report to comply with Standard VI(A) Disclosure of Conflicts.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #5 of 120

Question ID: 1214673

Fran Lester, CFA, works for a broker based in a country in which participation in any IPO is permitted with her employer's permission. She lives and works in a country that has no restrictions on her participation in IPOs. If Lester's firm is distributing shares of an oversubscribed IPO through the office Lester works in, can Lester receive shares in the IPO?

A) **No, not under any circumstances.**

B) Yes, but she must obtain permission from her employer.

C) Yes, because the applicable law is that of her home country.

Explanation

Standard I(A) Knowledge of the Law requires members and candidates to comply with the strictest requirement among the law where they reside, the law in the area where they do business, and the Code and Standards. In this case, the Code and Standards is the strictest. Standard III(B) Fair Dealing prohibits members and candidates from withholding shares in oversubscribed IPOs from clients for their own benefit.

For Further Reference:

(Study Session 1, Module 3.1 and 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #6 of 120

Question ID: 1214675

Diane Harris, a CFA Institute member, is a portfolio manager for Worldwide Investments. One of her clients has offered her the use of his condominium in Hawaii if the returns on his U.S. equities account beat their benchmark on a risk-adjusted basis. Harris informed her manager of all terms of this agreement in writing and received verbal consent to the arrangement before accepting the offer. Did Harris violate the Standards?

A) **Yes, because written consent from her employer is required.**

B) No, because bonuses from clients for doing her job well do not create a conflict of interest.

C) No, because Harris notified and received verbal consent from her employer to enter the arrangement.

Explanation

Standard IV(B) Additional Compensation Arrangements requires that members and candidates obtain written consent from their employers to enter into the agreement for additional contingent compensation from a client. Harris only received verbal consent. The concern is that such an arrangement might induce Harris to give partiality to this client's portfolio over those of her other clients.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #7 of 120

Question ID: 1214682

Four years ago, American Securities discontinued its international small-cap investment strategy and terminated the GIPS composite for presenting its performance. To comply with GIPS, does American Securities still need to offer a compliant presentation for this composite?

A) **Yes.**

B) No, because the composite was terminated more than three years ago.

C) No, because the composite does not reflect an investment strategy that the firm offers to clients.

Explanation

To comply with GIPS, firms must make compliant presentations available to prospects, on request, for any composite the firm has offered in the past five years, even for composites that have been terminated.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.d)

CFA® Program Curriculum, Volume 1, page 239

Question #8 of 120

Question ID: 1214677

Betty Cantor, CFA, has been finishing a report on high-tech firm HLC Corporation and intends to give it a "market perform" rating. Before releasing the report, she speaks with Donald Watson, her former manager and mentor, who is now with another investment firm and is well known for his great calls on high-tech companies. Watson is bullish on HLC and tells Cantor that "HLC is going to increase for sure over the next year, and we have it as a buy." Cantor changes her rating on HLC to "outperform" based on Watson's comments. Cantor has:

A) not violated the Standards because she knows the high quality of Watson's analysis.

B) violated the Standards because she based her rating on material nonpublic information.

C) **violated the Standards because she does not have a reasonable basis for her recommendation.**

Explanation

Cantor violated Standard V(A) Diligence and Reasonable Basis because she did not have a reasonable and adequate basis for recommending clients buy HLC. Watson's enthusiasm and claim that HLC stock will perform well are not consistent with Cantor's conclusions. She must perform her own analysis before changing her investment recommendation. She has also violated Standard I(B) Independence and Objectivity because she has not exhibited the independence of analysis and thought that is required by the Standard.

For Further Reference:

(Study Session 1, Module 3.1, 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #9 of 120

Question ID: 1214678

Tom Hayes, CFA, changed firms recently, becoming the senior analyst at Balcom Management. He had earned a great reputation at his old firm with his analysis of Selldex, which doubled in value after his recommendation. Because he still likes Selldex, Hayes recreates from public sources the records and analysis he did at his previous employer and issues a report on Selldex with a "buy" rating. Hayes has:

A) **not violated the Standards.**

B) violated the Standards because the analysis he did for his previous employer belongs to his previous employer.

C) violated the Standards because his analysis is based on previous research and not independent of his prior analysis.

Explanation

The actions are not in violation of the Standards. Under Standard IV(A) Loyalty, the use in new employment of knowledge and experience gained in previous employment is not prohibited. Hayes recreated the supporting records as required by Standard V(C) Record Retention.

For Further Reference:

(Study Session 1, Module 3.6, 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #10 of 120

Question ID: 1214681

The CFA Institute Code of Ethics *most likely* requires members and candidates to:

A) not engage in activity which compromises the integrity of CFA Institute.

B) stay informed on applicable laws and regulations that pertain to their respective areas of business.

C) **act with competence, integrity, and in ethical manner when dealing with the public, clients, employers, employees, and other market participants.**

Explanation

The first requirement of the Code of Ethics is that members and candidates "act with integrity, competence, diligence, respect, and in an ethical manner with the public, clients, prospective clients, employers, employees, colleagues in the investment profession, and other participants in the global capital markets." Knowing the applicable laws and regulations is required by Standard I(A) Knowledge of the Law. Standard VII(A) Conduct as Participants in CFA Institute Programs requires members and candidates not to compromise the integrity of CFA Institute.

For Further Reference:

(Study Session 1, Module 3.1, 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #11 of 120

Question ID: 1214676

Craig Boone, CFA, a fixed-income trader, observes that one of the salesmen on the desk has been allocating his trades at the end of the day, giving better execution to large clients, a practice Boone suspects is illegal. The salesman tells Boone this is a common practice and that the firm's senior management is aware of it. If Boone makes a personal record of the activity, takes it home for his personal files, and subsequently reveals it to regulatory authorities, he would:

A) **not be in violation of any Standards.**

B) be in violation of the Standards for disclosing confidential information.

C) be in violation of the Standards for breaching his duty of loyalty to his employer.

Explanation

According to Standard IV(A) Loyalty, the interests of a member or candidate's employer are secondary to protecting the interests of clients and the integrity of capital markets. In this circumstance, whistleblowing is justified. As long as his motivation is clearly not for personal gain, he may, according to the Standards, violate employer confidentiality in this case. While he is required to dissociate from the suspect activity by Standard I(A) Knowledge of the Law, he is not prohibited by the Standards from reporting it unless a stricter local law applies.

For Further Reference:

(Study Session 1, Module 3.1, 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #12 of 120

Question ID: 1214669

Lynn Black, CFA, is an analyst with the underwriter for an upcoming issue of Mtex Software debentures. Black learns from an employee in Mtex's programming department that there is a serious problem with Mtex's newest software program and that many customers have canceled their orders with Mtex. There is no mention of these problems in the prospectus for the debentures, which has been circulated. According to the CFA Institute Standards of Professional Conduct, Black's *best* course of action is to:

A) **inform her supervisor of her discovery.**

B) take no action because this is material nonpublic information.

C) notify potential investors of the omission on a fair and equitable basis.

Explanation

Black is in possession of material nonpublic information, and her most appropriate course of action is to inform her supervisor (or the firm's compliance officer) of what she has learned. She may not simply share the information with prospective buyers; if her firm determines that the information affects the value of the debentures, they must revise and recirculate the prospectus. Failing to do so may violate Standard I(C) Misrepresentation. Not informing her employer may be detrimental to her firm's interests and reputation if proceeding with the new issue without disclosure would violate regulations or laws.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #13 of 120

Question ID: 1214667

Sanctions that may be imposed on members by CFA Institute include:

A) public censure, suspension of membership, and fines.

B) suspension of membership, revocation of CFA charter, and fines.

C) **public censure, suspension of membership, and revocation of CFA charter.**

Explanation

If a Professional Conduct Program inquiry finds that a member has violated the Code and Standards, CFA Institute may impose sanctions, which include public censure, suspension of membership in CFA Institute and use of the CFA designation, and revocation of a member's CFA charter. CFA Institute does not impose fines.

For Further Reference:

(Study Session 1, Module 2.1, LOS 2.a)

CFA® Program Curriculum, Volume 1, page 39

Question #14 of 120

Question ID: 1214680

Mary Walters, CFA, is a bank trust officer who has entered into a referral agreement with Bob Sear, a tax attorney. Sear has told Walters that he will do her tax work in return for referrals. According to the CFA Institute Code and Standards, Walters must disclose:

A) only the fact that she compensated for referrals, to any clients or prospects she refers to Sear.

B) only the fact that she is compensated for referrals, to her employer and any clients or prospects she refers to Sear.

C) **the fact that she is compensated for the referrals and the nature of the compensation she is to receive, to her employer and any clients or prospects she refers to Sear.**

Explanation

Standard VI(C) Referral Fees requires members and candidates to disclose any compensation received for referrals and the nature of the compensation, to their employers and to any clients or prospects they refer to others with the expectation of compensation in any form.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #15 of 120

Question ID: 1214670

Ronaldo Jenkins, CFA, chief investment officer for Windwatch Advisors, has been helping his local municipality find an investment bank for a bond issue. Jenkins was told in confidence that one investment bank, which is a subsidiary of a commercial bank held in Windwatch client portfolios, is experiencing financial difficulties and will be shut down soon. According to the CFA Institute Standards of Professional Conduct, Jenkins is *least likely* permitted to:

A) share the information received about the investment bank with his compliance officer.

B) **share the information received about the investment bank with Windwatch's head of equity investments.**

C) approach the investment bank about making public disclosure of the financial difficulties and pending closure.

Explanation

The information received by Jenkins is covered by Standard II(A) Material Nonpublic Information, under which members who possess material nonpublic information related to the value of a security are prohibited from trading, or causing others to trade in, that security. For purposes of compliance with Standard II(A), Jenkins may attempt to achieve public dissemination of the information and may inform his compliance officer of the information. However, under no circumstances should Jenkins share material nonpublic information with other investment personnel. Sharing such information may cause others to trade on the information in violation of Standard II(A).

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #16 of 120

Question ID: 1214674

Morton Crane, CFA, is a portfolio manager. Crane has just been informed by his compliance officer that a new law will require additional disclosures of personal client information to regulators for two of Crane's former clients and one of his current clients. Crane decides to comply with the new law and provide the required client information. Has Crane violated CFA Institute Standards of Professional Conduct?

A) **No.**

B) Yes, because he disclosed confidential information about a former client.

C) Yes, because he disclosed confidential information about a current client.

Explanation

According to Standard III(E) Preservation of Confidentiality, members and candidates must keep information about former, current, and potential future clients confidential unless client information is legally required to be disclosed, the information pertains to potential illegal client activities, or the client gives permission for the information to be disclosed. Crane is complying with current legal reporting requirements, which require disclosure of personal client information for both former and current clients. Thus, Crane has not violated the Standard by disclosing the client information.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #17 of 120

Question ID: 1214671

While visiting Cassori Company, Mark Ramsey, CFA, overhears management make comments that are not public information but are not really meaningful by themselves. Combining this information with his own analysis and other outside sources, Ramsey decides to change his recommendation on Cassori from "Buy" to "Sell." According to the CFA Institute Standards of Professional Conduct, Ramsey:

A) must not issue his report until Cassori’s management makes their comments public.

B) **may issue his “Sell” report because the facts are nonmaterial, but should maintain a file of the facts and documents leading to this conclusion.**

C) must report these events to his immediate supervisor and legal counsel, since they have become material in combination with his analysis.

Explanation

According to Standard II(A) Material Nonpublic Information, the use of security analysis combined with nonmaterial nonpublic information to arrive at significant conclusions is allowable under the mosaic theory.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #18 of 120

Question ID: 1214683

Compliance with the Global Investment Performance Standards *least likely* requires firms to:

A) **include all fee-paying and non-fee-paying accounts in at least one composite.**

B) document in writing the policies and procedures used to comply with GIPS.

C) specifically define what constitutes the firm that is claiming compliance.

Explanation

All fee-paying discretionary accounts must be included in composites, but including non-fee-paying accounts is not required. The other statements are requirements for compliance.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.a)

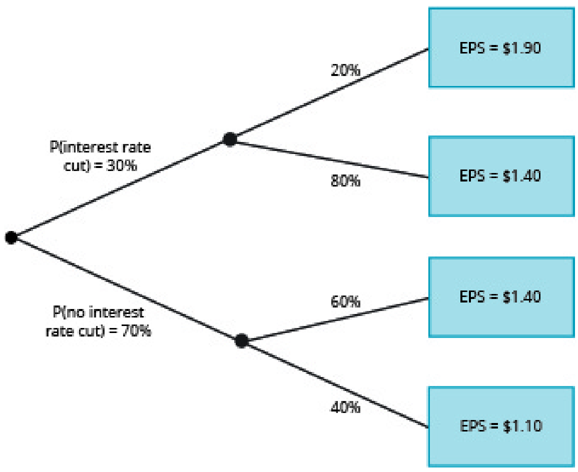
CFA® Program Curriculum, Volume 1, page 235

Question #19 of 120

Question ID: 1214691

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

Based on the tree diagram below:



The expected earnings per share is *closest* to:

A) $1.33.

B) **$1.35.**

C) $1.37.

Explanation

Expected EPS = (0.3)(0.2)1.90 + (0.3)(0.8)1.40 + (0.7)(0.6)1.40 + (0.7)(0.4)1.10 = $1.346.

For Further Reference:

(Study Session 2, Module 8.2, LOS 8.h, 8.j)

CFA® Program Curriculum, Volume 1, page 470

CFA® Program Curriculum, Volume 1, page 475

Question #20 of 120

Question ID: 1214686

An investor holds a portfolio consisting of one share of each of the following stocks:

Stock	Price at the Beginning of the Year	Price at the End of the Year	Cash Dividend During the Year
A	$10	$20	$0
B	$50	$60	$1
C	$100	$110	$4

For the 1-year holding period, the portfolio total return is *closest* to:

A) 15.79%.
B) 18.42%.
C) **21.88%.**

Explanation

$$\frac{195}{160} - 1 = 0.21875, \text{ or } 21.88\%$$

A longer approach is to calculate the holding period return for each stock for the year, then weight the returns using beginning values.

$$\begin{aligned}\text{Stock A} &= \frac{20-10}{10} = 100\% \\ \text{Stock B} &= \frac{60+1-50}{50} = 22\% \\ \text{Stock C} &= \frac{110+4-100}{100} = 14\%\end{aligned}$$

$$\left(\frac{10}{160}\right)(100\%) + \left(\frac{50}{160}\right)(22\%) + \left(\frac{100}{160}\right)(14\%) = 0.06250 + 0.06875 + 0.08750 = 0.21875, \text{ or } 21.88\%$$

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.e)

CFA® Program Curriculum, Volume 1, page 386

Question #21 of 120

Question ID: 1214685

Which of the following measurement scales provides the *least* information?

A) Ratio.

B) Ordinal.

C) **Nominal.**

Explanation

From least to most information, the ordering of measurements scales is nominal, ordinal, interval, and ratio.

For Further Reference:

(Study Session 2, Module 7.1, LOS 7.a)

CFA® Program Curriculum, Volume 1, page 371

Question #22 of 120

Question ID: 1214687

An analyst observes the following four annual returns: R_1 = +10%, R_2 = –15%, R_3 = 0%, and R_4 = +5%. The average compound annual rate over the four years is *closest* to:

A) –5.0%.

B) **–0.5%.**

C) 0.0%.

Explanation

$G = [(1.10)(0.85)(1.00)(1.05)]^{0.25} – 1$

$G = (0.98175)^{0.25} – 1 = 0.9954 – 1 = –0.00459 \approx –0.5\%$

Note: Taking a number to the 0.25 power is the same as taking the fourth root of the number.

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.e)

CFA® Program Curriculum, Volume 1, page 386

Question #23 of 120

Question ID: 1214688

Given the observations 45, 20, 30, and 25, the mean absolute deviation is *closest* to:

A) 0.0.

B) **7.5.**

C) 13.3.

Explanation

The mean absolute deviation is the mean of the absolute values of the differences between each number and the mean. Because the mean is (45 + 20 + 30 + 25) / 4 = 30, the mean absolute deviation is $\frac{15+10+0+5}{4} = 7.5$.

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.g)

CFA® Program Curriculum, Volume 1, page 411

Question #24 of 120

Question ID: 1214692

An investor wants to buy a condominium in Florida. The value of her portfolio is currently $1,000,000, and she needs $100,000 in one year for the down payment. She doesn't mind decreasing her capital as long she has $950,000 remaining in her portfolio after the down payment is made. She is considering two new portfolios for her holdings. The details on the two portfolios are:

	Expected Annual Return	Standard Deviation of Returns
Portfolio 1	17%	15%
Portfolio 2	12%	10%
Portfolio 3	8%	6%

According to Roy's Safety-First criterion, the portfolio she would prefer is:

A) **Portfolio 1.**

B) Portfolio 2.

C) Portfolio 3.

Explanation

As long as the investor earns at least a 5% return over the next year, the value of her portfolio after deducting the $100,000 down payment will not fall below $950,000. We first calculate the SFRatio for each of the two possible portfolios. We know that:

$$\text{SFRatio} = \frac{[\text{E}(\text{R}_\text{p}) - \text{R}_\text{L}]}{\sigma_\text{p}}$$

$$\text{SFRatio}_1 = \frac{17-5}{15} = 0.8$$

$$\text{SFRatio}_2 = \frac{12-5}{10} = 0.7$$

$$\text{SFRatio}_3 = \frac{8-5}{6} = 0.5$$

The SFRatio is the number of standard deviations that the minimum return is below the mean (expected) return. The optimal portfolio is the one with the greatest SFRatio as it has the lowest probability of a return below the minimum. Based on the SFRatio, the investor would prefer Portfolio 1.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.m)

CFA® Program Curriculum, Volume 1, page 537

Question #25 of 120

Question ID: 1214684

An investor would like to purchase a 20-year annuity that will pay $45,000 each year beginning when she retires 15 years from now. The amount she would need to invest today to fund this annuity, assuming an annual return of 5%, in order fund this annuity, is *closest* to:

A) $269,750.

B) $274,340.

C) **$283,240.**

Explanation

The amount needed to fund the annuity at t=15 is $588,839. In BGN mode: N = 20; I/Y = 5; PMT = 45,000; FV = 0; CPT PV. The equivalent amount at t=0 is $588,839 / $(1.05)^{15}$ = $283,242.

For Further Reference:

(Study Session 2, Module 6.3, LOS 6.e)

CFA® Program Curriculum, Volume 1, page 316

Question #26 of 120

Question ID: 1214693

Adam Farman has been asked to estimate the volatility of a technology stock index. He has identified a statistic which has an expected value equal to the population volatility and has determined that increasing his sample size will decrease the sampling error for this statistic. Based only on these properties, his statistic can *best* be described as:

A) unbiased and efficient.

B) **unbiased and consistent.**

C) efficient and consistent.

Explanation

An unbiased estimator has an expected value equal to the true value of the population parameter. A consistent estimator is more accurate the greater the sample size. An efficient estimator has the sampling distribution that is less than that of any other unbiased estimator.

For Further Reference:

(Study Session 3, Module 10.1, LOS 10.g)

CFA® Program Curriculum, Volume 1, page 580

Question #27 of 120

Question ID: 1214689

According to Chebyshev's inequality, the minimum proportion of observations falling within three standard deviations of the mean for a negatively skewed distribution is *closest* to:

A) 68%.

B) 75%.

C) **89%.**

Explanation

According to Chebyshev's inequality, the proportion of the observations within 3 standard deviations of the mean is at least $1 – (1 / 3^2)$ = 0.89 or 89%. This holds for any distribution, regardless of the shape.

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.h)

CFA® Program Curriculum, Volume 1, page 420

Question #28 of 120

Question ID: 1214690

The following table summarizes the results of a poll taken of CEOs and analysts about the economic impact of a pending piece of legislation:

Group	Number of Respondents who Predict Positive Impact	Number of Respondents who Predict Negative Impact
CEOs	40	30
Analysts	70	60

What is the probability that a randomly selected individual from this group will be either an analyst or someone who thinks this legislation will have a positive impact on the economy?

A) 0.75.

B) 0.80.

C) **0.85.**

Explanation

Using the addition rule for probabilities, P(analyst or positive) = P(analyst) + P(positive) − P(analyst and positive). P(A or positive) = 130 / 200 + 110 / 200 − (70 / 200) = 0.65 + 0.55 − 0.35 = 0.85. Alternatively, CEOs that predict positive impact = 40, analysts = 130, Prob (CEO positive or analyst) = (40 + 130) / 200 = 85%.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.f)

CFA® Program Curriculum, Volume 1, page 463

Question #29 of 120

Question ID: 1214694

A researcher has investigated the returns over the last five years to a long-short strategy based on mean reversion in equity returns volatility. His hypothesis test led to rejection of the hypothesis that abnormal (risk-adjusted) returns to the strategy over the period were less than or equal to zero at the 1% level of significance. He would *most* appropriately decide that:

A) his firm should employ the strategy for client accounts because the abnormal returns are positive and statistically significant.

B) **while the abnormal returns are highly significant statistically, they may not be economically meaningful.**

C) as long as the estimated statistical returns are greater than the transactions costs of the strategy, his firm should employ the strategy for client accounts.

Explanation

There are many reasons that a statistically significant result may not be economically significant (meaningful). Besides transactions costs, we must consider the risk of the strategy as well. For example, although the mean abnormal return to the strategy over the 5-year sample period is greater than transactions costs, abnormal returns for various sub-periods may be highly variable. In this case the risk of the strategy return from month to month or quarter to quarter may be too great to make employing the strategy in client accounts economically attractive.

For Further Reference:

(Study Session 3, Module 11.2, LOS 11.e)

CFA® Program Curriculum, Volume 1, page 619

Question #30 of 120

Question ID: 1214695

Which of the following tests would generally be considered a nonparametric test?

A) **Whether a sample is random or not.**
B) Large sample test of the value of a population mean.
C) Value of the variance of a normal population.

Explanation

Nonparametric tests can be used in a variety of instances where the assumptions required for parametric tests cannot be sustained. A runs test can be used to test for the randomness of a sample. Both of the other tests are parametric because they test the value of a parameter of the underlying distribution.

For Further Reference:

(Study Session 3, Module 11.3, LOS 11.l)

CFA® Program Curriculum, Volume 1, page 632

Question #31 of 120

Question ID: 1214705

Questions 31 through 42 relate to Economics. (18 minutes)

Assume the exchange rate between the Trotter (TRT) and the Roeckl (RKL) is 5.50 TRT/RKL and the exchange rate between the Roeckl and the Passage (PSG) is 8.00 RKL/PSG. The cross rate between the PSG and the TRT is *closest* to:

A) **0.0227 PSG/TRT.**
B) 0.6875 PSG/TRT.
C) 44.00 PSG/TRT.

Explanation

The TRT/PSG cross rate is 5.5 × 8.0 = 44 TRT/PSG. Because the answer choices are quoted as PSG/TRT, we need to invert this result: 1 / 44 = 0.0227 PSG/TRT.

For Further Reference:

(Study Session 5, Module 18.1, LOS 18.d)

CFA® Program Curriculum, Volume 2, page 417

Question #32 of 120

Question ID: 1214699

Which of the following statements about methods of calculating gross domestic product is *most accurate*?

- **A)** **Except for a statistical discrepancy, the income and expenditure approaches to calculating GDP should result in the same value for economic output.**
- **B)** Because it includes activity at all stages of production, the sum-of-value-added method results in a better estimate of GDP than the value-of-final-output method.
- **C)** Value-of-final-output is used to calculate GDP under the expenditure approach, while sum-of-value-added is used to calculate GDP under the income approach.

Explanation

Because aggregate income is the same as aggregate output, measuring GDP by summing incomes or expenditures should produce the same value, except for a statistical discrepancy that results from using different data sources. The sum-of-value-added method of calculating GDP records the sum of the increases in value of goods and services at each stage of their production and distribution. The resulting total for GDP is the same as that reached by the value-of-final-output method because the sum of value added to a good at all stages of processing is equal to its selling price. Both methods calculate GDP based on expenditures.

For Further Reference:

(Study Session 4, Module 14.1, LOS 14.a, 14.b)

CFA® Program Curriculum, Volume 2, page 118

CFA® Program Curriculum, Volume 2, page 119

Question #33 of 120

Question ID: 1214706

The J-curve, in the context of trade between two countries, refers to the fact that when the domestic country has a trade deficit:

- **A)** appreciation of the domestic currency initially leads to a decrease in the trade deficit but will increase the trade deficit in the long term.
- **B)** an increase in domestic inflation will initially increase the trade deficit but will decrease the trade deficit in the long term.
- **C)** **appreciation of the foreign currency will initially increase the trade deficit but will decrease the trade deficit in the long term.**

Explanation

The J-curve effect refers to a plot of the trade deficit over time when the domestic currency depreciates (the foreign currency appreciates). The trade deficit gets worse initially but then improves over time, either because export and import demand are more elastic in the long run or because existing contracts for future delivery are fixed in foreign currency terms in the short run.

For Further Reference:

(Study Session 5, Module 18.3, LOS 18.j)

CFA® Program Curriculum, Volume 2, page 440

Question #34 of 120

Question ID: 1214703

Which of the following organizations is the most focused on promoting economic growth and reducing poverty by offering both monetary and technical assistance?

A) **World Bank.**
B) World Trade Organization.
C) International Monetary Fund.

Explanation

Promoting economic growth and reducing world poverty are among the primary goals of the World Bank. The IMF primarily promotes the growth of international trade, supports exchange rate stability, and provides a forum for cooperation on monetary problems internationally. The WTO has a primary focus on reaching trade agreements and settling trade disputes.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.j)

CFA® Program Curriculum, Volume 2, page 377

Question #35 of 120

Question ID: 1214701

The quantity theory of money states that in a full employment economy, any increase in the supply of money in excess of the rate of growth of real GDP will lead to a proportional increase in:

A) **the price level.**
B) velocity.
C) real GDP.

Explanation

The quantity theory of money hypothesizes that a change in the money supply, at full employment, will cause a proportional change in the price level because velocity and real output will be unaffected. According to the equation of exchange, MV = PY, output of goods and services produced, *Y*, at full employment cannot change, so the price level, *P*, must increase.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.c)

CFA® Program Curriculum, Volume 2, page 264

Question #36 of 120

Question ID: 1214707

Akor is a country that has chosen to use a conventional fixed peg arrangement as the country's exchange rate regime. Under this arrangement, Akor's exchange rate against the currency to which it pegs:

A) is market-determined.
B) will be equal to the peg rate.
C) **may fluctuate around the peg rate.**

Explanation

In a conventional fixed peg arrangement, a country pegs its currency within a margin of ±1% versus another currency or a basket that includes the currencies of its major trading or financial partners. Market-determined exchange rates are a characteristic of an independently floating exchange rate regime.

For Further Reference:

(Study Session 5, Module 18.3, LOS 18.i)

CFA® Program Curriculum, Volume 2, page 428

Question #37 of 120

Question ID: 1214696

A plant manager observes the following relationship between hours of labor and units of output:

Labor hours	Units of output
1,400	140,000
1,500	150,000
1,600	165,000
1,700	175,000
1,800	180,000
1,900	170,000

At what quantity of labor hours does this plant first experience diminishing marginal returns to labor?

A) Between 1,400 and 1,500 hours.
B) **Between 1,600 and 1,700 hours.**
C) Between 1,800 and 1,900 hours.

Explanation

Diminishing marginal returns occur when an additional unit of an input will produce a smaller gain in output than the previous unit of the input. In this case, marginal returns are increasing with up to 1,600 hours of labor input and begin decreasing between 1,600 and 1,700 hours of labor input.

For Further Reference:

(Study Session 4, Module 12.2, LOS 12.d)

CFA® Program Curriculum, Volume 2, page 23

Question #38 of 120

Question ID: 1214698

Which of the following statements about a monopolist is *most accurate*? A monopolist will:

A) maximize the average profit per unit sold.
B) charge the highest price for which it can sell its product.
C) **produce where marginal revenue equals marginal cost.**

Explanation

Like all price searchers, monopolists will expand output until marginal revenue equals marginal cost. Monopolists do not charge the highest possible price which would be the price resulting in only one sale. A monopolist seeks to maximize profit, not price.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.d)

CFA® Program Curriculum, Volume 2, page 64

Question #39 of 120

Question ID: 1214704

Which of the following objectives would a national government *most likely* pursue by placing restrictions on inflows of foreign capital?

A) Protecting domestic industries.
B) Supporting domestic asset prices.
C) Keeping domestic interest rates low.

Explanation

National governments may seek to limit foreign investment in certain domestic industries, for example, those that are seen as essential for national defense. A government seeking to keep domestic interest rates low or support domestic asset prices would be more likely to restrict capital outflows.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.g)

CFA® Program Curriculum, Volume 2, page 362

Question #40 of 120

Question ID: 1214702

Under what conditions is inflation *most likely* to shift wealth from lenders to borrowers?

A) Only when inflation is unexpected.
B) Only when inflation is unexpected and negative.
C) When inflation is either unexpected or expected.

Explanation

Inflation that is unexpected, or higher than expected, shifts wealth from lenders to borrowers. Unexpected deflation has the opposite effect. Because interest rates include a premium for expected inflation, an inflation rate that matches expectations does not shift wealth from lenders to borrowers.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.g)

CFA® Program Curriculum, Volume 2, page 279

Question #41 of 120

Question ID: 1214697

The short-run supply curve for a firm under perfect competition is the firm's:

A) marginal cost curve above average total cost.

B) **marginal cost curve above average variable cost.**

C) average variable cost curve above marginal revenue.

Explanation

The supply curve for a firm under perfect competition is its marginal cost curve above average variable cost. As long as price exceeds AVC, the firm will produce up to the quantity where MC = Price, which is also MR in this case.

For Further Reference:

(Study Session 4, Module 13.1, LOS 13.c)

CFA® Program Curriculum, Volume 2, page 64

Question #42 of 120

Question ID: 1214700

Long-run aggregate supply is *least likely* to be affected by changes in the:

A) **prices of raw materials inputs.**

B) quantity of labor in the economy.

C) level of technology.

Explanation

Long-run aggregate supply is related to the level of technology and the available quantities of labor and capital. If the prices of productive inputs increase, short-run aggregate supply decreases (the SRAS curve shifts to the left), but long-run aggregate supply (potential real GDP) is unaffected.

For Further Reference:

(Study Session 4, Modules 14.2, 14.3, LOS 14.h, 14.m)

CFA® Program Curriculum, Volume 2, page 148

CFA® Program Curriculum, Volume 2, page 172

Question #43 of 120

Question ID: 1214709

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

In accrual accounting, the matching principle states that:

A) an entity should recognize revenues only when received and expenses only when they are paid.

B) transactions and events producing cash flows are allocated only to time periods in which the cash flows occur.

C) **expenses incurred to generate revenue are recognized in the same time period as the revenue.**

Explanation

The matching principle holds that expenses should be accounted for in the same performance measurement period as the revenue they generate.

For Further Reference:

(Study Session 7, Module 21.3, LOS 21.d)

CFA® Program Curriculum, Volume 3, page 84

Question #44 of 120

Question ID: 1214712

Listed equity securities held as assets that do not convey significant influence in the investee company must be reported at fair value through profit and loss under:

A) IFRS only.

B) **U.S. GAAP only.**

C) both IFRS and U.S. GAAP.

Explanation

U.S. GAAP categorizes equity investment without significant control as trading securities, reported at fair value with profit and loss reported on the income statement. Under IFRS, firms can report equity securities in this manner, but may elect at the time of purchase to report an equity security at fair value through other comprehensive income.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #45 of 120

Question ID: 1214708

The independent auditors of Shadydells, Inc., have determined that the company recorded operating leases that should have been capitalized. This is considered to be a material instance of noncompliance with applicable GAAP, but the financial statements are otherwise fairly presented. Which opinion are the auditors *most likely* to issue?

A) Adverse opinion.

B) **Qualified opinion.**

C) Unqualified opinion.

Explanation

Auditors issue a qualified opinion when there is a material instance of noncompliance with applicable accounting standards. An unqualified opinion means the auditors believe there is reasonable assurance that the financial statements are free of error and in compliance with applicable accounting standards. An adverse opinion is only issued when the auditors believe the financial statements as a whole are not fairly presented.

For Further Reference:

(Study Session 6, Module 19.2, LOS 19.d)

Question #46 of 120

Question ID: 1214714

An analyst gathers the following information:

Net income	$100
Decreased in accounts receivable	30
Depreciation	25
Increase in inventory	17
Increase in accounts payable	10
Decrease in wages payable	5
Increase in deferred taxes	17
Sale of fixed assets	150
Purchase of fixed assets	340
Profit from sale of fixed assets	5
Dividends paid out	35
Sale of new common stock	120

Based on the above information, the company's cash flow from operations under U.S. GAAP is:

A) **$155.**
B) $165.
C) $182.

Explanation

Net income	+100
Adjustment for noncash and nonoperating items:	
Depreciation	+25
Deferred taxes (increase)	+17
Profit from sale of equipment	–5
Adjustment for working capital items:	
Accounts receivable (decrease)	+30
Inventory (increase)	-17
Accounts payable (increase)	+10
Wages payable (decrease)	–5
Cash flow from operations	+$155

Dividends paid are CFF, not CFO.

For Further Reference:

(Study Session 7, Module 23.2, LOS 23.f)

CFA® Program Curriculum, Volume 3, page 199

Question #47 of 120

Question ID: 1214713

Which of the following statements about cash flow is *least accurate*? Under U.S. GAAP, cash flow from:

A) operations includes cash operating expenses and changes in working capital accounts.

B) financing includes the proceeds of debt issued and from the sale of the company's common stock.

C) **investing includes interest income from investment in debt securities.**

Explanation

Interest income is considered an operating cash flow under U.S. GAAP.

For Further Reference:

(Study Session 7, Module 23.1, LOS 23.a)

CFA® Program Curriculum, Volume 3, page 185

Question #48 of 120

Question ID: 1214710

An analyst gathered the following data about a company:

- The company had 500,000 shares of common stock outstanding for the entire year.
- The company's beginning stock price was $40, its ending price was $60, and its average price over the year was $50.
- The company has 120,000 warrants outstanding for the entire year.
- Each warrant allows the holder to buy one share of common stock at $45 per share.

How many shares of common stock should the company use in computing its diluted earnings per share?

A) 488,000.

B) 500,000.

C) **512,000.**

Explanation

Dilution occurs since the exercise price for the warrants ($45) is less than the average market price for the shares ($50). The incremental number of shares outstanding is found from:

$$\left(\frac{\text{market price}-\text{exercise price}}{\text{market price}}\right) \times \#\text{ warrants} = \left(\frac{50-45}{50}\right) \times 120{,}000 = 12{,}000$$

Number of shares to use in diluted EPS calculation = 500,000 + 12,000 = 512,000.

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g, 21.h)

CFA® Program Curriculum, Volume 3, page 99

CFA® Program Curriculum, Volume 3, page 99

Question #49 of 120

Question ID: 1214716

Books Forever, Inc., uses short-term bank debt to buy inventory. Assuming an initial current ratio that is greater than 1, and an initial quick (or acid test) ratio that is less than 1, what is the effect of these transactions on the current ratio and the quick ratio?

A) **Both ratios will decrease.**
B) Neither ratio will decrease.
C) Only one ratio will decrease.

Explanation

As an example, start with CA = 2, CL = 1, and Inv = 1.2. We begin with a current ratio of 2 and a quick ratio of 0.8. If the firm increases short-term bank debt (a current liability) by 1 to buy inventory (a current asset) of 1, both the numerator and denominator increase by 1, resulting in $\frac{3}{2} = 1.5$ (new current ratio) and $\frac{3-2.2}{2} = 0.4$ (new quick ratio).

For Further Reference:

(Study Session 7, Module 24.2, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #50 of 120

Question ID: 1214723

Under U.S. GAAP, a lessor will recognize the lease value as revenue and the carrying value of the asset as cost of goods sold if the lease is classified as:

A) an operating lease.
B) **a sales-type lease.**
C) a direct financing lease.

Explanation

Under U.S. GAAP, a lessor will recognize the lease value as revenue and the carrying value of the asset as cost of goods sold if the lease is classified as a sales-type lease.

For Further Reference:

(Study Session 8, Module 28.4, LOS 28.h)

CFA® Program Curriculum, Volume 3, page 511

Question #51 of 120

Question ID: 1214711

Which of the following types of items are *least likely* included in other comprehensive income?

A) Losses under the revaluation model.
B) Unrealized gains on equity securities.

C) Realized gains on securities classified as available-for-sale.

Explanation

Realized gains on investment securities are reported on the income statement under both IFRS and U.S. GAAP. The other two types of gains may be reported as other comprehensive income under certain circumstances.

For Further Reference:

(Study Session 7, Module 21.5, LOS 21.k)

CFA® Program Curriculum, Volume 3, page 113

Question #52 of 120

Question ID: 1214718

Which of the following accounting practices is *most likely* to decrease reported earnings in the current period?

A) Using the straight-line method of depreciation instead of an accelerated method.

B) Capitalizing advertising expenses rather than expensing them in the current period.

C) Using LIFO inventory cost methods during a period of rising prices.

Explanation

LIFO will result in lower net income than FIFO in the current period, during a period of rising prices. The other choices will tend to increase current period earnings.

For Further Reference:

(Study Session 8, Modules 25.1, 26.1, 26.2, LOS 25.c, 26.c, 26.e)

CFA® Program Curriculum, Volume 3, page 316

CFA® Program Curriculum, Volume 3, page 386

CFA® Program Curriculum, Volume 3, page 404

Question #53 of 120

Question ID: 1214725

Which of the following pairs of general categories are *least likely* to be considered in the formulas used by credit rating agencies to determine the capacity of a borrower to repay a debt?

A) Operational efficiency; leverage.

B) Margin stability; availability of collateral.

C) Leverage; scale and diversification.

Explanation

The four general categories are: (1) scale and diversification, (2) operational efficiency, (3) margin stability, and (4) leverage. Larger companies and those with more different product lines and greater geographic diversification are better credit risks. High operating efficiency is indicative of a better credit risk. Stable profit margins indicate a higher probability of repayment and thus, a better credit risk. Firms with greater earnings in relation to their debt level are better credit risks. While the availability of collateral certainly reduces lender risk, it is not one of the general categories used by credit rating agencies to determine *capacity* to repay. Specifically, they would consider (1) several specific accounting ratios and (2) business characteristics. The availability of collateral falls into neither category.

For Further Reference:

(Study Session 9, Module 30.2, LOS 30.c)

CFA® Program Curriculum, Volume 3, page 622

Question #54 of 120

Question ID: 1214715

A common-size cash flow statement is *least likely* to show each cash inflow as a percentage of:

A) revenue.
B) all cash inflows.
C) **total cash flows.**

Explanation

Common-size cash flow statements show each cash flow item as a percentage of revenue or show each cash flow outflow as a percentage of all cash outflows and each cash inflow as a percentage of all cash inflows.

For Further Reference:

(Study Session 7, Module 23.4, LOS 23.i)

CFA® Program Curriculum, Volume 3, page 222

Question #55 of 120

Question ID: 1214724

During a period when net income is unexpectedly weak, managers who attempt to smooth earnings are *most likely* to:

A) **capitalize an expense.**
B) capitalize a new lease.
C) classify a nonrecurring gain as recurring income.

Explanation

Management may attempt to increase reported earnings in the current period by capitalizing an expense. Capitalizing a lease would decrease earnings in the current period compared to recording an operating lease. Classifying a nonrecurring gain as recurring income would not increase net income because it already includes nonrecurring gains.

For Further Reference:

(Study Session 9, Module 29.2, LOS 29.h)

CFA® Program Curriculum, Volume 3, page 573

Question #56 of 120

Question ID: 1214720

Which of the following definitions used in accounting for income taxes is *least accurate*?

A) Income tax expense is current period taxes payable adjusted for any changes in deferred tax assets and liabilities.

B) A valuation allowance is a reserve against deferred tax assets based on the likelihood that those assets will not be realized.

C) **A deferred tax liability is created when tax expense is less than taxes payable and the difference is expected to reverse in future years.**

Explanation

Deferred tax liability refers to balance sheet amounts that are created when tax expense is greater than taxes payable.

For Further Reference:

(Study Session 8, Module 27.1, LOS 27.a)

CFA® Program Curriculum, Volume 3, page 452

Question #57 of 120

Question ID: 1214722

Under U.S. GAAP, which of the following statements about the financial statement effects of issuing bonds is *least accurate*?

A) Issuance of debt has no effect on cash flow from operations.

B) Periodic interest payments decrease cash flow from operations by the amount of interest paid.

C) **Payment of debt at maturity decreases cash flow from operations by the face value of the debt.**

Explanation

Issuing debt results in a cash inflow from financing. Payment of debt at maturity has no effect on cash flow from operations but decreases cash flow from financing by the face value of the debt.

For Further Reference:

(Study Session 8, Modules 28.1, 28.2, LOS 28.a, 28.b)

CFA® Program Curriculum, Volume 3, page 490

CFA® Program Curriculum, Volume 3, page 494

Question #58 of 120

Question ID: 1214721

When an increase in the tax rate is enacted, deferred tax:

A) **assets and liabilities both increase in value.**

B) assets decrease in value and deferred tax liabilities increase in value.

C) liability and asset accounts are maintained at historical tax rates until they reverse.

Explanation

The liability method (SFAS 109 of U.S. GAAP) takes a balance sheet approach and adjusts deferred tax assets and liabilities to future tax rates. An increase in the tax rate increases the value of both deferred tax assets and deferred tax liabilities.

For Further Reference:

(Study Session 8, Module 27.4, LOS 27.e)

CFA® Program Curriculum, Volume 3, page 461

Question #59 of 120

Question ID: 1214719

Under U.S. GAAP, an asset is considered impaired if its book value is:

A) less than its market value.

B) greater than the present value of its expected future cash flows.

C) **greater than the sum of its undiscounted expected cash flows.**

Explanation

Under U.S. GAAP, an asset is considered impaired when its book value is greater than the sum of the estimated undiscounted future cash flows from its use and disposal.

For Further Reference:

(Study Session 8, Module 26.3, LOS 26.i)

CFA® Program Curriculum, Volume 3, page 411

Question #60 of 120

Question ID: 1214717

In periods of rising prices and stable or increasing inventory quantities, compared with companies that use LIFO inventory accounting, companies that use the FIFO method will have:

A) higher COGS and lower taxes.

B) **higher net income and higher taxes.**

C) lower inventory balances and lower working capital.

Explanation

FIFO companies have higher net income, lower COGS, higher inventory, and higher taxes.

For Further Reference:

(Study Session 8, Module 25.5, LOS 25.k, 25.l)

CFA® Program Curriculum, Volume 3, page 340

CFA® Program Curriculum, Volume 3, page 340

Question #61 of 120

Question ID: 1214730

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

Which of the following statements about the component costs of capital is *least accurate*?

A) The cost of common equity is the required rate of return on common stock.

B) **The cost of preferred stock is the preferred dividend divided by the preferred's par value.**

C) The after-tax cost of debt is based on the expected yield to maturity on newly issued debt.

Explanation

The cost of preferred stock is calculated as the preferred dividend divided by the market price, not the par value.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.g)

CFA® Program Curriculum, Volume 4, page 86

Question #62 of 120

Question ID: 1214729

Project X has an internal rate of return (IRR) of 14%. Project Y has an IRR of 17%. Both projects have conventional cash flow patterns (all inflows after the initial cash outflow). If the required rate of return is:

A) **14%, the net present value (NPV) of Project Y will exceed the NPV of Project X.**

B) greater than 17%, Project Y will have a shorter payback period than Project X.

C) 10%, both projects will have a positive NPV, and the NPV of Project Y will exceed the NPV of Project X.

Explanation

We know that at a 14% discount rate, the NPV of Project X is zero and the NPV of Project Y is greater than zero. There is no well-defined relationship between the required rate of return and ordinary payback. If Project Y is smaller in size, its NPV may be smaller than that of Project X.

For Further Reference:

(Study Session 10, Modules 32.1, 32.2, LOS 32.d, 32.e)

CFA® Program Curriculum, Volume 4, page 52

CFA® Program Curriculum, Volume 4, page 58

Question #63 of 120

Question ID: 1214728

AlcoBanc owns a piece of property that is under consideration for a new bank branch. Which of the following is *least likely* a relevant incremental cash flow in analyzing a capital budgeting project? The:

A) **interest costs of a loan for the property purchase.**

B) business gained at other branches due to new customers at the proposed site.

C) $150,000 AlcoBanc could get if they sold the property instead of building a new branch.

Explanation

Financing costs should not be included in incremental cash flows. They are reflected in the weighted average cost of capital (WACC). New business at other branches is a positive externality and the $150,000 to sell the property is an opportunity cost.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.b)

CFA® Program Curriculum, Volume 4, page 50

Question #64 of 120

Question ID: 1214732

Jay Construction Company is considering whether to accept a new bridge-building project. Jay will use the pure-play method to estimate the cost of capital for the project, using Cass Bridge Builders as a comparable company. To calculate the project beta, Jay should:

A) estimate Cass's cost of equity capital and apply it to the project.

B) use the CAPM equation, substituting Cass's equity beta for its own.

C) **adjust Cass's equity beta for any difference in leverage between Cass and Jay.**

Explanation

To use the pure-play method, an asset beta is calculated by removing the effects of leverage (delevering) from the comparable company's equity beta, then a project beta is estimated by adjusting the asset beta (relevering) based on the capital structure of the company that is evaluating the project.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.i)

CFA® Program Curriculum, Volume 4, page 94

Question #65 of 120

Question ID: 1214735

Which of the following would *most likely* indicate deterioration of a firm's working capital management?

A) An increase in days of payables outstanding.

B) **An increase in days of receivables outstanding.**

C) A decreased amount of cash and cash equivalents.

Explanation

An increase in days of receivables outstanding, other things equal, will lengthen both the operating and cash conversion cycles, indicating poorer working capital management. An increase in days of payables outstanding, other things equal, would decrease the cash conversion cycle. A decrease in cash and marketable securities could simply indicate better management of cash (e.g., buying back its common stock or investing excess cash in profitable business opportunities or securities).

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.c)

CFA® Program Curriculum, Volume 4, page 159

Question #66 of 120

Question ID: 1214743

Under which of the following shareholder voting structures is a minority shareholder group *most likely* to achieve representation on a board of directors?

A) Majority voting.

B) **Cumulative voting.**

C) Supermajority voting.

Explanation

In a board election for multiple seats, cumulative voting permits shareholders to cast their total number of votes (number of shares owned times the number of board seats being voted on) for a single candidate. This makes it easier for a minority shareholder group to gain board representation compared to majority voting, in which shareholders must vote for each board seat separately. A supermajority refers to the percentage of votes that may be required to pass a special resolution.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.e and Study Session 13, Module 39.1, LOS 39.b)

CFA® Program Curriculum, Volume 4, page 14

CFA® Program Curriculum, Volume 4, page 358

Question #67 of 120

Question ID: 1214733

Pannonia Enterprises, Inc. (PEI) has a target capital structure of 40% debt with 60% equity. PEI's pretax cost of debt will remain at 9% until the firm raises more than $200,000 in new debt capital, at which point its pretax cost of debt will increase to 9.5%. PEI's cost of equity will increase when more than $400,000 in equity capital is raised. PEI's break point for debt capital is *closest* to:

A) $200,000.

B) **$500,000.**

C) $666,667.

Explanation

$$\text{capital component breakpoint} = \frac{\text{value at which component's cost of capital changes}}{\text{component's weight in WACC}}$$

$$\text{debt breakpoint} = \frac{\$200{,}000}{0.40} = \$500{,}000$$

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.k)

CFA® Program Curriculum, Volume 4, page 102

Question #68 of 120

Question ID: 1214734

For a profitable company, issuing debt in order to retire common stock will *most likely*:

A) increase both net income and return on equity.

B) decrease both operating income and net income.

C) **increase both the level and variability of return on equity.**

Explanation

An increase in debt will increase interest expense, which will decrease net income but not operating income, which is calculated before subtracting interest expense. For a profitable firm, the decrease in net income will be offset by the decrease in equity from the repurchase of common stock, so that ROE increases. The effect of the increase in financial leverage will, however, increase the variability of ROE for a given change in operating earnings.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.c)

CFA® Program Curriculum, Volume 4, page 137

Question #69 of 120

Question ID: 1214726

With respect to firms in the financial services industry, a conflict between the interests of shareholders and regulators is *best* illustrated by a shareholder preference for:

A) a decrease in the firm's equity capital.
B) a decrease in the firm's equity multiplier.
C) an increase in the value of a firm's equity.

Explanation

The shareholders of a regulated firm in the financial services industry may prefer to maintain less equity capital to support their business to increase the return on equity, while regulators may impose higher equity capital requirements to protect the economic system in cases of financial distress. Regulators are not typically concerned with increasing equity values but rather with a firm's accounting equity. A decrease in the equity multiplier (financial leverage ratio, or assets/equity) will result from an increase in a firm's equity capital.

For Further Reference:

(Study Session 7, Module 24.4, LOS 24.d and Study Session 10, Module 31.1, LOS 31.b)

CFA® Program Curriculum, Volume 3, page 279

CFA® Program Curriculum, Volume 4, page 8

Question #70 of 120

Question ID: 1214731

Given the following information on a company's capital structure and a marginal tax rate of 40%:

Type of Capital	Weight in Capital Structure	Before-Tax Component Cost
Bonds	40%	7.5%
Preferred stock	5%	11%
Common stock	55%	15%

The company's weighted average cost of capital is *closest to*:

A) 10.0%.
B) 10.6%.
C) 11.8%.

Explanation

WACC = $(w_d)(k_d)(1 – t) + (w_{ps})(k_{ps}) + (w_{ce})(k_{ce})$ = (0.4)(7.5)(1 – 0.4) + (0.05)(11) + (0.55)(15) = 10.6%.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.a)

CFA® Program Curriculum, Volume 4, page 78

Question #71 of 120

Question ID: 1214736

McDermott Company routinely takes advantage of trade discounts on terms of 2/15 net 90. The cost of short-term borrowing for McDermott is 10.2%. If McDermott begins forgoing trade discounts and paying invoices on their due dates, this would *most likely*:

A) increase its cost of short-term funding.

B) improve its business relationships with its suppliers.

C) be viewed by analysts as efficient accounts payable management.

Explanation

Forgoing the discount can be viewed as borrowing funds from suppliers for 90 – 15 = 75 days. Given trade credit terms of 2/15 net 90, the annualized cost of this funding is:

$$\left(1+\frac{0.02}{1-0.02}\right)^{\frac{365}{75}}-1=10.33\%$$

Because this source of financing is more costly than the company's other sources of short-term funding (10.2%), using it would not be viewed as efficient accounts payable management. Paying later is unlikely to improve its business relationship with its suppliers, although it will not necessarily have a negative impact on the relationship.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.f)

CFA® Program Curriculum, Volume 4, page 175

Question #72 of 120

Question ID: 1214727

Which of the following statements about corporate governance is *most accurate*?

A) Activist shareholders are likely to favor a policy of staggered board elections.

B) Effective control functions are likely to improve a company’s operating and financial performance.

C) Dual class structures are used to align the interests of the board with the interests of the majority of shareholders.

Explanation

Effective control functions can lead to better financial performance by reducing the risks and costs associated with weak controls. Staggered board elections make it more difficult for activist shareholders to gain control of a board of directors. Dual-class structures may allow a shareholder group, such as company founders or their heirs, to maintain control of the company even though they own less than 50% of the firm's common equity.

For Further Reference:

(Study Session 10, Module 31.2, LOS 31.g, 31.h, 31.i)

CFA® Program Curriculum, Volume 4, page 25

CFA® Program Curriculum, Volume 4, page 28

CFA® Program Curriculum, Volume 4, page 31

Question #73 of 120

Question ID: 1214738

Questions 73 through 85 relate to Equity Investments. (19.5 minutes)

A financial market is said to be operationally efficient if:

A) transactions costs are low.
B) prices adjust quickly to new information.
C) it results in resources being allocated to their most productive uses.

Explanation

Financial markets are said to be operationally efficient if participants can achieve their desired purposes with low transactions costs. Markets are said to be informationally efficient if prices reflect new information rapidly, and allocationally efficient if resources are allocated to their most productive uses.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.k)

CFA® Program Curriculum, Volume 4, page 260

Question #74 of 120

Question ID: 1214741

The weak form of the efficient market hypothesis (EMH) implies that:

A) no one can achieve abnormal returns using market information.
B) insiders, such as specialists and corporate board members, cannot achieve abnormal returns on average.
C) investors cannot achieve abnormal returns, on average, using technical analysis, after adjusting for transaction costs and taxes.

Explanation

The weak form of the EMH implies that an investor cannot earn positive abnormal returns on average using technical analysis (market information), after adjusting for transaction costs and taxes. Evidence has shown that insiders can achieve positive abnormal returns on average, but this relates to the strong form of the EMH.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

CFA® Program Curriculum, Volume 4, page 330

Question #75 of 120

Question ID: 1214739

Which of the following statements about alternative investment indexes is *most accurate*?

A) An investor can replicate a commodity index by making direct investments in the underlying physical commodities.

B) **Real Estate Investment Trust indexes track the prices of shares of publicly traded companies that invest in mortgages or real property.**

C) Hedge fund indexes accurately represent the investment performance of the hedge fund industry.

Explanation

REIT indexes represent a convenient way to invest in real estate. Commodity indexes are based on futures prices of commodities, and are not replicated by investing in the commodities themselves. Hedge fund indexes are biased upward because hedge funds are not required to disclose their performance to index providers and poorly performing funds are less likely to do so (self-selection bias).

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.j)

CFA® Program Curriculum, Volume 4, page 303

Question #76 of 120

Question ID: 1214745

The present value of an equity security's future cash flows is *most likely* to be significantly different from the company's:

A) **book value per share.**

B) market value per share.

C) intrinsic value per share.

Explanation

A company's book value per share is typically not equal to its intrinsic value per share, since book value is based on the historical cost of assets. Intrinsic value is the present value of the security's future cash flows. The market value of a share is the cumulative investors' estimation of this intrinsic value, and market value is usually somewhat in line with the intrinsic value.

For Further Reference:

(Study Session 13, Module 39.2, LOS 39.g)

CFA® Program Curriculum, Volume 4, page 373

Question #77 of 120

Question ID: 1214747

An analyst is valuing a company's perpetual preferred stock that pays a $6 annual dividend. The company's bonds currently yield 7.5% and preferred shares are selling to yield 75 basis points below the company's bond yield. The value of the preferred stock is *closest* to:

A) $72.

B) $80.

C) **$89.**

Explanation

Step 1:	Determine the discount rate. 7.50% – 0.75% = 6.75%.
Step 2:	Value the preferred. $V_P = \dfrac{\text{dividend}}{k_p} = \dfrac{\$6.00}{0.0675} = \$88.89$.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.f)

CFA® Program Curriculum, Volume 4, page 456

Question #78 of 120

Question ID: 1214737

An investor buys a stock at $32 a share and deposits 50% initial margin. Assume that the maintenance margin is 25%, the stock pays no dividends, and transaction costs and interest on the margin loan are zero. The price at which the investor would receive a margin call is *closest* to:

A) $16.

B) **$21.**

C) $24.

Explanation

Price at which a margin call will occur:

original price × [(1 − initial margin) / (1 − maintenance margin)]

= [$32.00 × (1 − 0.5)] / (1 − 0.25) = $16.00 / 0.75 = $21.33

For Further Reference:

(Study Session 12, Module 36.2, LOS 36.f)

CFA® Program Curriculum, Volume 4, page 243

Question #79 of 120

Question ID: 1214748

The risk-free rate is 5%, and the expected return on the market index is 15%. A stock has a:

- Beta of 1.0.
- Dividend payout ratio of 40%.
- Return on equity (ROE) of 15%.

If the stock is expected to pay a $2.50 dividend, its intrinsic value using dividend discount model is *closest* to:

A) $27.77.

B) **$41.67.**

C) $53.33.

Explanation

$E(R) = R_f + \text{beta}(R_M - R_f)$

$k = E(R) = 0.05 + 1(0.15 - 0.05) = 0.15$

Retention (b) = (1 − dividend payout ratio) = 1 − 0.4 = 0.6

g = (ROE)(b) = (0.15)(0.6) = 0.09

$$\text{Value} = \frac{D_1}{k-g} = \frac{\$2.50}{0.15-0.09} = \$41.67$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #80 of 120

Question ID: 1214746

Which of the following industries is *best* described as non-cyclical and defensive?

A) Energy.

B) Technology.

C) **Consumer staples.**

Explanation

The consumer staples industry is best classified as non-cyclical and defensive. Energy and technology are best classified as cyclical industries.

For Further Reference:

(Study Session 13, Module 40.1, 40.2, LOS 40.c, 40.i)

CFA® Program Curriculum, Volume 4, page 390

CFA® Program Curriculum, Volume 4, page 420

Question #81 of 120

Question ID: 1214742

Which of the following scenarios is inconsistent with efficient financial markets?

A) An analyst's buy recommendations have returned 2% more than the broad market index, on average.

B) Johnson, Inc. reports an increase of 8% in its earnings from a year earlier, and its stock price declines 5% on the news.

C) **Earl Baker, an investor, earns consistently superior risk-adjusted returns by buying stocks when most investment advisors are pessimistic and selling stocks when most investment advisors are optimistic.**

Explanation

If markets are efficient, investors should not earn consistently superior returns from technical trading rules such as a contrary opinion strategy. A stock price will decrease on a report of an increase in earnings if that increase is less than investors expected based on all the information previously available. An analyst's recommendations could be for stocks with more than market risk, or the analyst's industry may outperform for some period of time. Both could account for outperformance even though markets are efficient.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

CFA® Program Curriculum, Volume 4, page 330

Question #82 of 120

Question ID: 1214744

The holder of the type of security that has a priority in liquidation less than that of bonds or promissory notes issued by the company but ahead of that of common stock is *most likely* to have:

A) **no voting rights.**
B) statutory voting rights.
C) cumulative voting rights.

Explanation

The security being described here is preferred stock (preference shares). In the event of liquidation of a firm, the claims of preferred equity shareholders are senior to the claims of common stockholders but subordinated to the claims of debt holders. Preferred shares typically have no voting rights.

For Further Reference:

(Study Session 13, Module 39.1, LOS 39.a)

CFA® Program Curriculum, Volume 4, page 357

Question #83 of 120

Question ID: 1214740

Transactions costs incurred from portfolio rebalancing are *most likely* to be highest for funds that track:

A) price-weighted indexes.
B) value-weighted indexes.
C) **equal-weighted indexes.**

Explanation

Equal-weighted index portfolios require rebalancing after each return period because differences in returns among securities will drive the security weightings away from equal weighting. Changes in the security prices automatically adjust the weights in price- and value-weighted index portfolios to their correct values, so funds that track price- and value-weighted indexes do not require frequent rebalancing.

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.f)

CFA® Program Curriculum, Volume 4, page 293

Question #84 of 120

Question ID: 1214749

A stock that currently does not pay a dividend is expected to pay its first dividend of $1.00 five years from today. Thereafter, the dividend is expected to grow at an annual rate of 25% for the next three years and then grow at a constant rate of 5% per year thereafter. The required rate of return is 10.3%. The value of the stock today is *closest* to:

- **A) $20.65.**
- **B)** $22.72.
- **C)** $23.87.

Explanation

This is essentially a two-stage dividend discount model (DDM) problem. Discounting all future cash flows, we get:

$$P_0 = \frac{1.00}{(1.103)^5} + \frac{1.25}{(1.103)^6} + \frac{(1.25)^2}{(1.103)^7} + \left[\frac{(1.25)^3}{(0.103-0.05)(1.103)^7}\right] = \$20.647$$

Note that the constant growth formula can be applied to dividend 8 (1.25^3) because it *will* grow at a constant rate (5%) forever.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #85 of 120

Question ID: 1214750

Roger Gould, CFA, is analyzing the stock of Zero Incorporated and trying to determine which price multiple to use in his valuation. Zero, a start-up, had losses over the last 12 months and is projected to have a loss over the next 12 months. Zero has significant internally generated intangible assets and human capital that Gould would like to capture in his valuation. The price multiple that Gould should *most appropriately* use is the:

- **A) price-to-sales ratio.**
- **B)** price-to-book value.
- **C)** price-to-earnings ratio.

Explanation

The price-to-sales ratio can be used for firms with negative earnings and implicitly recognizes the value of assets not recognized on the balance sheet. The price-to-earnings ratio is the most popular ratio in the investment community; however, it is not a useful ratio for firms with negative earnings. One of the significant disadvantages of the price-to-book value ratio is that it does not recognize the value of nonphysical assets such as human capital or intangible assets generated internally. Therefore, price-to-book value would not be the best choice.

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.j, 41.l)

SchweserNotes, Book 4 page 215

CFA® Program Curriculum, Volume 4, page 468

CFA® Program Curriculum, Volume 4, page 468

Question #86 of 120

Question ID: 1214757

Questions 86 through 99 relate to Fixed Income. (21 minutes)

Asset-backed securities (ABS) may have a higher credit rating than the seller's corporate bonds because:

A) **they are issued by a special purpose entity.**
B) the seller's ABS are senior to its corporate bonds.
C) ABS are investment grade while corporate bonds may be speculative grade.

Explanation

The SPE in a securitization is bankruptcy-remote from the seller, which means the seller's creditors do not have a claim against the pool of assets underlying an ABS. As a result, the ABS may have a higher credit rating than the seller's corporate bonds.

For Further Reference:

(Study Session 14, Module 45.1, LOS 45.b)

CFA® Program Curriculum: Volume 5, page 181

Question #87 of 120

Question ID: 1214759

All else equal, which of the following is *least likely* to increase the interest rate risk of a bond?

A) A longer maturity.
B) **Inclusion of a call feature.**
C) A decrease in the YTM.

Explanation

Inclusion of a call feature will decrease the duration of a fixed income security. The other choices increase duration.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.e)

CFA® Program Curriculum: Volume 5, page 259

Question #88 of 120

Question ID: 1214760

An investor with an investment horizon of 5 years has purchased a 15-year 6% coupon bond at par. The bond has a modified duration of 9.8. The duration gap for this investor is *closest* to:

A) –4.2.
B) 4.8.
C) **5.4.**

Explanation

The duration gap is calculated as the bond's Macaulay duration minus the investor's investment horizon. Modified duration is Macaulay duration / (1 + YTM), so we can calculate the Macaulay duration as 9.8 × 1.06 = 10.388. The duration gap is 10.388 – 5 = 5.388.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.k)

CFA® Program Curriculum: Volume 5, page 279

Question #89 of 120

Question ID: 1214762

Which of the following assets is *most likely* to represent high-quality collateral in credit analysis?

A) Goodwill.

B) **Trademarks.**

C) Deferred tax assets.

Explanation

Intangible assets that can be sold, such as trademarks, provide collateral of good quality. A credit analyst should view as low-quality collateral any assets that are likely to be written down in value if a firm encounters financial distress, such as goodwill and deferred tax assets.

For Further Reference:

(Study Session 15, Module 47.1, LOS 47.f)

CFA® Program Curriculum: Volume 5, page 321

Question #90 of 120

Question ID: 1214753

A company has two $1,000 face value bonds outstanding both selling for $701.22. The first issue has an annual coupon of 8% and 20 years to maturity. The second bond has the same yield to maturity as the first bond but has only five years remaining until maturity. The second issue pays interest annually as well. What is the annual interest payment on the second issue?

A) $18.56.

B) $27.18.

C) **$37.12.**

Explanation

First find the yield to maturity (YTM) of the first bond and use it in the second bond calculation. The calculator sequence to determine the YTM is: PV = –701.22; FV = 1,000; PMT = 80; N = 20; CPT → I/Y = 12.00%. We discount the cash flows of the second bond at 12.00%. The calculator steps are: PV = –701.22; FV = 1,000; N = 5; I/Y = 12; CPT → PMT = $37.12.

For Further Reference:

(Study Session 14, Module 44.1, LOS 44.a, 44.b)

CFA® Program Curriculum: Volume 5, page 108

CFA® Program Curriculum: Volume 5, page 113

Question #91 of 120

Question ID: 1214761

If a bond has a convexity of 120 and a modified duration of 10, the convexity adjustment (to a duration-based approximation) associated with a 25 basis point interest rate decline is *closest* to:

A) –2.875%.
B) –2.125%.
C) **+0.0375%.**

Explanation

$$\begin{aligned}\text{convexity adjustment to } \%\Delta\text{P} &= \frac{1}{2}\text{convexity measure} \times (\Delta\text{YTM})^2 \times 100 \\ &= \frac{1}{2}(120)(-0.0025)^2(100) \\ &= 0.0375\%\end{aligned}$$

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.i)

CFA® Program Curriculum: Volume 5, page 270

Question #92 of 120

Question ID: 1214764

Recent economic data suggest an increasing likelihood that the economy will soon enter a recessionary phase. What is the *most likely* effect on the yields of lower-quality corporate bonds and on credit spreads of lower-quality versus higher-quality corporate bonds?

A) **Both will increase.**
B) Both will decrease.
C) One will increase and one will decrease.

Explanation

During economic contractions, the probability of default increases for lower-quality issues and their yields increase. When investors anticipate an economic downturn, they tend to sell low-quality issues and buy high-quality issues, causing credit spreads to widen.

For Further Reference:

(Study Session 15, Module 47.2, LOS 47.i, 47.j)

CFA® Program Curriculum: Volume 5, page 338

CFA® Program Curriculum: Volume 5, page 347

Question #93 of 120

Question ID: 1214752

A supranational bond:

A) **is issued by a multilateral agency.**
B) trades outside the jurisdiction of any one country.
C) pays coupons in a different currency than its principal.

Explanation

Supranational bonds are issued by multilateral agencies such as the World Bank or International Monetary Fund. Bonds that trade outside the jurisdiction of any one country are known as global bonds. Bonds that pay coupons in a different currency than their principal are known as dual-currency bonds.

For Further Reference:

(Study Session 14, Modules 42.1, 43.1, LOS 42.a, 43.f)

CFA® Program Curriculum: Volume 5, page 7

CFA® Program Curriculum: Volume 5, page 77

Question #94 of 120

Question ID: 1214763

If an investor wants only investment grade bonds in her portfolio, she would be *least likely* to purchase a:

A) 2-year municipal bond rated A–.
B) **3-year municipal bond rated BB.**
C) 15-year, semiannual coupon corporate bond rated BBB.

Explanation

Investment grade bonds are BBB– and above. This bond is rated BB, which is below BBB–.

For Further Reference:

(Study Session 14, Module 43.1, LOS 43.a and Study Session 15, Module 47.1, LOS 47.d)

CFA® Program Curriculum: Volume 5, page 56

CFA® Program Curriculum: Volume 5, page 313

Question #95 of 120

Question ID: 1214754

If the yield curve is downward-sloping, the no-arbitrage value of a bond calculated using spot rates will be:

A) **equal to the market price of the bond.**
B) less than the market price of the bond.
C) greater than the market price of the bond.

Explanation

The value of a bond calculated using appropriate spot rates is its no-arbitrage value. If no arbitrage opportunities are present, this value is equal to the market price of a bond.

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.c)

CFA® Program Curriculum: Volume 5, page 117

Question #96 of 120

Question ID: 1214758

Changes in a fixed-coupon bond's cash flows that result from changes in yield would be reflected in the bond's:

A) **effective duration.**
B) modified duration.
C) Macaulay duration.

Explanation

Effective duration and effective convexity capture the effects from changes in a bond's cash flows when the yield changes (e.g., an increase in the likelihood of a bond being called when its yield has decreased). For this reason, they are the appropriate measures of interest rate sensitivity for bonds with embedded options.

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.b, 46.c)

CFA® Program Curriculum: Volume 5, page 248

CFA® Program Curriculum: Volume 5, page 255

Question #97 of 120

Question ID: 1214756

Four non-convertible bonds have the indicated yield spreads to Treasury securities:

	Maturity	Government Spread	Zero- Volatility Spread	Option- Adjusted Spread
Bond W	2 years	156 bp	155 bp	130 bp
Bond X	3 years	173 bp	174 bp	199 bp
Bond Y	5 years	188 bp	189 bp	164 bp
Bond Z	10 years	202 bp	201 bp	226 bp

Based on these spreads, it is *most likely* that:

A) Bond X is callable, and Bond Y is putable.
B) **Bond W is callable, and Bond Z is putable.**
C) Bond Z is callable, and the spot yield curve is inverted.

Explanation

Bonds W and Y are most likely callable, and Bonds X and Z are most likely putable. If the option-adjusted spread is less than the zero-volatility spread, the embedded option has a negative value to the bondholder (e.g., a call option), and if the option-adjusted spread is greater than the zero-volatility spread, the embedded option has a positive value to the bondholder (e.g., a put option). Zero-volatility spreads adjust for the fact that nominal spreads (between the yields to maturity of two bonds) are theoretically correct only when the spot yield curve is flat. All of these bonds' zero-volatility spreads are nearly identical to their government spreads, which suggests the spot yield curve is approximately flat.

For Further Reference:

(Study Session 14, Modules 42.2, 44.5, LOS 42.f, 44.k)

CFA® Program Curriculum: Volume 5, page 37

CFA® Program Curriculum: Volume 5, page 147

Question #98 of 120

Question ID: 1214755

Based on the following interest rates:

1-year spot rate	3.0%
1-year forward rate one year from now	5.0%
2-year forward rate one year from now	6.5%

The 3-year spot rate is *closest to*:

A) 5.0%.
B) **5.3%.**
C) 9.3%.

Explanation

The 3-year spot rate is the discount rate for a 3-year zero-coupon security. To eliminate any opportunity for arbitrage, the 3-year spot rate should equal the compounded equivalent of borrowing at expected forward rates:

$$\text{3-year spot rate} = [(1 + 0.03)(1 + 0.065)^2]^{1/3} - 1 = 0.0532 = 5.32\%.$$

Note that the answer can be approximated simply by averaging the 1-year rate and the 2-year forward rate one year from now: (3 + 6.5 + 6.5) / 3 = 5.33%.

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.j)

CFA® Program Curriculum: Volume 5, page 143

Question #99 of 120

Question ID: 1214751

A step-up coupon bond is structured such that its coupon rate increases:

A) **on a predetermined schedule.**
B) if a reference interest rate increases.
C) if the issuer’s credit rating decreases.

Explanation

Step-up coupon bonds feature a coupon rate that increases on a predetermined schedule. Credit linked coupon bonds have a coupon rate that changes inversely with the issuer's credit rating. Floating-rate notes have coupon rates that are based on a reference interest rate.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.e)

CFA® Program Curriculum: Volume 5, page 26

Question #100 of 120

Question ID: 1214769

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

Based on parity relations for options prices, the value of a put minus the value of a call (on the same asset with same exercise prices) must equal:

A) the price of a forward contract on the asset minus the present value of the option exercise price.

B) the present value of the price of a forward contract on the asset minus the option exercise price.

C) **the present value of the difference between the option exercise price and the price of a forward contract on the asset.**

Explanation

The no-arbitrage put-call-forward parity condition for European options can be presented as:

$$\frac{F_0(T)}{(1+R_f)^T} + p_0 = c_0 + \frac{X}{(1+R_f)^T}$$

or as

$$p_0 - c_0 = \frac{X - F_0(T)}{(1+R_f)^T}$$

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.m)

CFA® Program Curriculum, Volume 5, page 494

Question #101 of 120

Question ID: 1214767

The price of a forward contract:

A) **is stated in the contract.**

B) equals zero at contract initiation.

C) may fluctuate during the contract's life.

Explanation

The *price* of a forward contract is the price at which the underlying asset will be exchanged at settlement. This price is stated in the contract and determined such that the *value* of the contract is equal to zero at initiation. The value may vary during the life of the contract but the price does not.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.b, 49.d)

CFA® Program Curriculum, Volume 5, page 468

CFA® Program Curriculum, Volume 5, page 469

Question #102 of 120

Question ID: 1214765

Two Level I CFA candidates are discussing futures and make the following statements:

Candidate 1:	Futures are traded using standardized contracts. They require margin and incur interest charges on the margin loan.
Candidate 2:	If the margin balance falls below the maintenance margin amount due to a change in the contract price for the underlying assets, the investor must add funds to bring the margin back up to the initial margin requirement.

Are the candidates' statements correct or incorrect?

A) Both statements are correct.

B) Neither statement is correct.

C) **Only one of the statements is correct.**

Explanation

Candidate 1 is incorrect. In the futures markets, margin is a performance guarantee. It is money deposited by both the long and the short. There is no loan involved (contrasting with margin in the equity markets) and thus no interest charges. Candidate 2 is correct.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #103 of 120

Question ID: 1214766

Which of the following statements for puts at expiration is *least accurate*? The:

A) put buyer's maximum loss is the put option's premium.

B) maximum loss to the writer of a put is the strike price less the premium.

C) **put holder will exercise the option whenever the stock's price is greater than the exercise price.**

Explanation

The put holder will exercise the option at expiration whenever the stock's price is less than the exercise price.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.c, 48.d)

CFA® Program Curriculum, Volume 5, page 394

CFA® Program Curriculum, Volume 5, page 406

Question #104 of 120

Question ID: 1214768

At initiation of a forward contract and a futures contract with identical terms, their prices are *most likely* to be different if:

A) the spot price is highly volatile.

B) the forward contract is marked to market daily.

C) **short-term interest rates are negatively correlated with futures prices.**

Explanation

Forward and futures prices may differ for otherwise identical contracts if interest rates are positively or negatively correlated with futures prices. The difference arises from the fact that futures are marked to market daily while forwards are not. A futures contract holder can earn interest on mark-to-market gains and faces an opportunity cost of interest of mark-to-market losses.

For Further Reference:

(Study Session 16, Module 49.2, LOS 49.g)

CFA® Program Curriculum, Volume 5, page 476

Question #105 of 120

Question ID: 1214770

Other things equal, a decrease in the value of a put option on a stock is *most likely* consistent with which of the following changes in the risk-free rate and stock return volatility?

	Risk-free rate	Volatility
A)	**Increase**	**Decrease**
B)	Decrease	Increase
C)	Decrease	Decrease

Explanation

Decreasing volatility of returns on the underlying stock will decrease option values for both puts and calls. An increase in the risk-free rate increases the values of call options on equities and decreases the values of put options on equities.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 482

Question #106 of 120

Question ID: 1214771

An American put option on a stock is *most likely* to have a higher value than an otherwise identical European put option:

A) if the stock pays a dividend.

B) when the options are at the money.

C) **when the options are in the money.**

Explanation

For a put option, the right to exercise early can be valuable when the option is deep in the money. Therefore, an American put that is in the money will have a higher value than an otherwise identical European put. On their expiration date, American and European options have the same value because both can be exercised. Cash flows from the underlying asset do not make the right to exercise early more valuable for put options.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.o)

CFA® Program Curriculum, Volume 5, page 500

Question #107 of 120

Question ID: 1214772

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

Compared to the standard deviation of returns on a repeat sales index for a class of real estate properties, the standard deviation of returns on an appraisal index for the same class of properties is *most likely* to be:

A) lower.
B) higher.
C) the same.

Explanation

Appraisal index returns are based on estimates of property values and therefore tend to be smoother than repeat sales index returns, which are based on actual sale prices of properties. As a result, appraisal index returns have lower standard deviations.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #108 of 120

Question ID: 1214785

Compared to brownfield infrastructure investments, greenfield infrastructure investments are more appropriate for investors with:

A) lower liquidity needs and lower risk tolerance.
B) higher risk tolerance and shorter time horizons.
C) longer time horizons and lower liquidity needs.

Explanation

Greenfield investments are infrastructure projects that have not yet been built. Compared to brownfield investments, which are infrastructure assets that already exist, greenfield investments are more appropriate for investors with longer time horizons. Greenfield investments have greater risk than brownfield investments. Direct investments in infrastructure of either type are illiquid.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b and Study Session 19, Module 54.1, LOS 54.e)

CFA® Program Curriculum, Volume 6, page 10

CFA® Program Curriculum, Volume 6, page 257

Question #109 of 120

Question ID: 1214773

Which of the following is *least likely* a type of hedge fund strategy?

A) Event-driven.

B) Market-neutral.

C) **Exchange-traded.**

Explanation

Hedge funds do not typically trade on exchanges.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #110 of 120

Question ID: 1214774

Compared with purchasing commodities, long positions in commodity derivatives offer the benefit of:

A) **no storage costs.**

B) convenience yield.

C) better correlation with spot prices.

Explanation

While commodity futures retain the risk and correlation characteristics of the underlying commodities, the investor does not incur storage costs. Derivatives cannot have higher correlation with spot prices than the commodity itself, as its price is the spot price. Convenience yield is a benefit of owning the actual commodities.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #111 of 120

Question ID: 1214775

Returns on which of the following alternative investments are *most likely* to include a liquidity premium?

A) **Private equity funds.**

B) Commodity-linked ETFs.

C) Real estate investment trusts.

Explanation

Investors require a liquidity premium for investments that are illiquid. Private equity funds are typically illiquid and may require lockup periods. Real estate investment trusts and exchange-traded funds are publicly traded and usually highly liquid.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.f)

CFA® Program Curriculum, Volume 6, page 57

Question #112 of 120

Question ID: 1214776

The risk measure for alternative investments that is *least likely* to be affected by tendency of returns to be leptokurtic is:

A) **the Sortino ratio.**
B) standard deviation.
C) coefficient of variation.

Explanation

Standard deviation and coefficient of variation, which is based on standard deviation of returns, both may be misleading for alternative investments because their returns distributions tend to be leptokurtic (i.e., they have "fat tails"). The Sortino ratio, in contrast, measures risk as downside deviation from the mean.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.f)

CFA® Program Curriculum, Volume 6, page 57

Question #113 of 120

Question ID: 1214777

If a commodity market is in backwardation:

A) it produces a negative roll yield.
B) **futures contract prices are less than the spot prices.**
C) a long futures position will increase in value as contracts approach expiration.

Explanation

Backwardation means futures prices are less than spot prices or that futures contracts with later expiration dates are priced lower than contracts with earlier expiration dates. This results in a positive roll yield as futures prices converge to the spot prices. This does not necessarily increase the value of a long position because the spot price may decrease.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #114 of 120

Question ID: 1214781

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

An investment advisor constructs a portfolio that plots on the capital market line but has less risk and a lower return than the market portfolio. This portfolio is *most accurately* described as:

A) **a lending portfolio.**

B) a leveraged portfolio.

C) an inefficient portfolio.

Explanation

A portfolio that plots on the capital market line with less risk than the market portfolio must include a positive allocation to the risk-free asset. Such a portfolio is called a lending portfolio because investing in the risk-free asset represents lending at the risk-free rate.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.b)

CFA® Program Curriculum, Volume 6, page 192

Question #115 of 120

Question ID: 1214779

Smith has more steeply sloped risk-return indifference curves than Jones. Assuming these investors have the same expectations, which of the following *best* describes their risk preferences and the characteristics of their optimal portfolios? Smith is:

A) less risk averse than Jones and will choose an optimal portfolio with a lower expected return.

B) **more risk averse than Jones and will choose an optimal portfolio with a lower expected return.**

C) more risk averse than Jones and will choose an optimal portfolio with a higher expected return.

Explanation

Steeply sloped risk-return indifference curves indicate that a greater increase in expected return is required as compensation for assuming an additional unit of risk, compared to less-steep indifference curves. The more risk-averse Smith will choose an optimal portfolio with lower risk and a lower expected return than the less risk-averse Jones's optimal portfolio.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.i)

CFA® Program Curriculum, Volume 6, page 166

Question #116 of 120

Question ID: 1214784

A stock has a beta of 0.9 and an estimated return of 10%. The risk-free rate is 7%, and the expected return on the market is 11%. According to the CAPM, this stock:

A) **is overvalued.**

B) is undervalued.

C) is properly valued.

Explanation

$E(R) = 7\% + 0.9(11\% - 7\%) = 10.6\%$. Because the expected return of 10% is less than the required return of 10.6%, the security is overvalued.

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.h)

CFA® Program Curriculum, Volume 6, page 217

Question #117 of 120

Question ID: 1214778

In which step of the portfolio management process does an investment manager rebalance the portfolio to its target asset allocation percentages?

A) Analysis step.
B) **Feedback step.**
C) Execution step.

Explanation

The three major steps in the portfolio management process are planning, execution, and feedback. Rebalancing the portfolio to its desired asset allocation is part of the feedback step.

For Further Reference:

(Study Session 18, Module 51.1, LOS 51.d)

CFA® Program Curriculum, Volume 6, page 94

Question #118 of 120

Question ID: 1214780

Which of the following possible portfolios cannot lie on the efficient frontier?

Portfolio	Expected Return	Standard Deviation
1	8%	6%
2	10%	6%
3	14%	12%
4	14%	16%

A) Portfolio 3 only.
B) **Portfolios 1 and 4.**
C) Portfolios 2 and 3.

Explanation

Portfolio 1 does not lie on the efficient frontier because it has a lower return than Portfolio 2 but has equal risk. Portfolio 4 does not lie on the efficient frontier because it has higher risk than Portfolio 3 but has the same return.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.h)

CFA® Program Curriculum, Volume 6, page 163

Question #119 of 120

Question ID: 1214782

Which of the following terms refer to the same type of risk?

- **A)** **Total risk and the variance of returns.**
- **B)** Systematic risk and firm-specific risk.
- **C)** Undiversifiable risk and unsystematic risk.

Explanation

Variance is a measure of total risk.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.c)

CFA® Program Curriculum, Volume 6, page 204

Question #120 of 120

Question ID: 1214783

Greenbaum, Inc. stock pays no dividend and currently trades at $54. Based on the CAPM and assuming an expected return on the market of 12% and a risk-free rate of 8%, the expected price for Greenbaum one year from now is $62. The beta of Greenbaum shares is *closest* to:

- **A)** 1.5.
- **B)** 1.6.
- **C)** **1.7.**

Explanation

Greenbaum's expected return for the coming year is (62 / 54) − 1 = 14.8%. Its risk premium relative to the risk-free rate is 14.8% – 8.0% = 6.8%. The market risk premium is 12% − 8% = 4%. Therefore, the beta of Greenbaum must be 6.8 / 4.0 = 1.7.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.e)

CFA® Program Curriculum, Volume 6, page 206